

Basic Science

Jingshu Keli attenuates cervical spinal nerve ligation-induced allodynia in rats through inhibition of spinal microglia and Stat3 activation

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Abstract

BACKGROUND CONTEXT: Cervical radicular pain resulting from mechanical compression of a spinal nerve secondary to spinal degenerative alternations negatively impacts patients' quality of life. Jingshu Keli (JSKL), a traditional Chinese medicine formula with multiple active compounds, has been prescribed for pain management in patients with cervical radiculopathy for decades. Two major components of JSKL, ferulic acid and cinnamaldehyde, were identified to have anti-inflammation effect via inhibiting activation of Stat3.

PURPOSE: To investigate the efficacy of JSKL by investigating its mechanism in attenuating cervical radiculopathy-induced mechanical allodynia via modulation activation of spinal microglia and phosphorylation of signal transducer and activator of transcription 3 (Stat3).

STUDY DESIGN: An in vivo animal experiment.

METHODS: Cervical radiculopathy of rats was established by C7 spinal nerve ligation (SNL) with 6-0 silk suture. The effect of postoperative daily gavage of JSKL on mechanical allodynia of rats was tested on day 3, 7, and 14 after surgery. Furthermore, spinal glial cells activation and phosphorylation of Stat3 (p-Stat3) were tested with immunofluorescence imaging and Western blot.

RESULT: The JSKL significantly inhibited SNL-induced allodynia as well as microglia activation in the spinal cord on day 7 and 14 after surgery. Moreover, expression of p-Stat3 was decreased in rats with SNL and JSKL treatment in comparison with rats with SNL and vehicle treatment.

CONCLUSIONS: The JSKL attenuated SNL-induced mechanical allodynia in rats. This analgesic effect might be explained by the suppression of activations of spinal microglia as well as p-Stat3. Our study provides experimental evidence for JSKL as an alternative approach to manage refractory pain in patients with cervical radiculopathy. © 2018 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license.

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Cervical radiculopathy; Mechanical allodynia; Microglia; Spinal nerve ligation; Stat3; Traditional Chinese Medicine JSKL.

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Introduction

Cervical radiculopathy greatly affects patients' quality of life with symptoms of upper limb pain or motor dysfunctions [1]. The annual incidence was reported to be 107.3 per 10,000 for men and 63.5 per 10,000 for women [2]. It has been shown that neurogenic inflammation induced by mechanical compression of cervical spinal nerve root is the major culprit contributing to radicular pain [1]. Currently, nonsteroid anti-inflammatory drugs, opioids, and steroids are prescribed as conservative treatments [1,3–5]. However, pain often remains insufficiently controlled and further investigations aiming better treatment approaches is warranted [6,7].

Both efficacy and underlying mechanisms of traditional herbal medicine formulated for treating human diseases, especially musculoskeletal disorders have been tested scientifically in the recent years [8–13]. Jingshu Keli (JSKL) is such a traditional herbal formula that has been prescribed for treating radiculopathy pain in the past few decades in China [14]. Yet, it lacks preclinical research evidence elucidating its cellular and molecular mechanism of action. The JSKL contains concentrated herbal granules made of extracts from *Notoginseng Radix*, *Angelicae Sinensis Radix*, *Chuanxiong Rhizoma*, *Carthami Flos*, *Gastrodiae Rhizoma*, *Cinnamomi Cortex*, and *Bovis Calculus Artifactus* [14], where ferulic acid from *Angelicae Sinensis Radix* and *Chuanxiong Rhizoma*, and cinnamaldehyde from *Cinnamomi Cortex*, which are two major biologically active components of JSKL, were reported to partially downregulate inflammatory mediators via preventing the activation of signal transducer and activator of transcription 3 (Stat3) in vitro [15,16]. On the other hand, spinal glial cells activation and the resulting neuroinflammation have been proven to give rise to neuropathic pain [17]. We hereby hypothesized that JSKL could attenuate spinal nerve ligation-induced pain symptom through modulating glial cell activation and inhibiting phosphorylation of Stat3 in a validated rat model of cervical radiculopathy.

Methods and materials

Animals

All animal experiments were approved by the Animal Experimentation Ethic Committee of the Chinese University of Hong Kong (Reference number: 15-170-MIS). Male SD rats weighing 180 to 250 g were obtained from the animal centre of CUHK. The rats were housed on a 12-12-hour dark-light cycle at 25°C and allowed free access to water and food.

Surgical procedure

An established procedure of cervical spinal nerve ligation (SNL) was applied to induce forepaw allodynia in rats [18,19]. Briefly, a longitudinal incision was made on

the back of rat's neck, followed by dissection of para spinal muscles to expose T2 spinous process. The right C7 spinal nerve was exposed via hemilaminectomy and removal of transverse process at the C6 and C7 levels. Then a tight ligation was made of the C7 spinal nerve distal to the dorsal root ganglion with 6-0 silk thread (Fig. 1). Animals assigned to the sham group received the same procedures except for SNL. The wound was closed by 5-0 silk thread.

Drug administration

Rats were assigned to three groups: sham operation with PBS administration (sham), SNL with PBS administration (SNL and vehicle), and SNL with JSKL administration (SNL and JSKL). The JSKL (provided by Sinopharm Group Jingfang Pharmaceutical Co., Ltd., Anhui, PR China). In clinical settings, oral JSKL is prescribed at a dose of 300 mg/kg/day to treat cervical radiculopathy, which is equivalent to 400 mg/day for rat weighing 180 to 250 g in our study based on human-animal drug dose conversion formula (animal equivalent dose = human dose * Km ratio) described previously [14,20,21]. In addition, our pilot study showed that JSKL at a dose of 400 mg/day led to better analgesic effect in comparison with lower dose (100 and 200 mg/day) on postoperative day (POD) 7. Therefore, in our current study JSKL was dissolved in phosphate-buffered saline (PBS) with a final concentration of 200 mg/ml and daily administered to rat with 2 ml of JSKL solution (400 mg/day/rat), or PBS by intragastric gavage from POD1 to 14 [20,22].

Mechanical allodynia

Eight rats from the sham group, 12 from vehicle, and 12 from JSKL group were tested for mechanical allodynia before surgery (baseline), and on POD 3, 7 and 14. Von Frey filaments were applied according to our established protocol [23]. Briefly, animals were put on the testing mesh 30 minutes before experiment for acclimation. A series of von Frey filaments were applied to the forepaw plantar surface. Positive responses were defined as withdrawing, licking or spasm of the forepaw. Three rounds of tests were

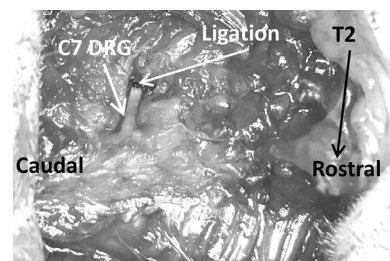


Fig. 1. Schematic description of surgical procedure. Ligation of the C7 spinal nerve was performed with 6-0 silk thread at the nerve root distal to dorsal root ganglion (DRG), after hemilaminectomy to expose the spinal cord and afferent nerve. T2 spinous process was exposed as the anatomic landmark to locate C7 vertebral disc.

applied to each rat at each time point. The paw withdrawal threshold (g) was calculated using a previously published method [24].

Immunofluorescence staining

Spinal microglia and astrocytes were visualized by staining with ionized calcium binding adaptor molecule 1 (Iba-1) and glial fibrillary acidic protein (GFAP), respectively. C7 spinal segments were harvested after transcardiac perfusion with 0.9% saline and 4% paraformaldehyde on POD 7 and 14. The spinal cord was postfixed in 4% paraformaldehyde for 24 hours and dehydrated in 30% sucrose for 48 hours before optimal cutting temperature compound embedding. Transverse sections with a thickness of 25 μ m was prepared. Sections were masked with blocking buffer (5% FBS and 0.03% Triton X-100 in PBS) for one hour and incubated overnight with rabbit anti-Iba-1 (1:1000, Wako, Japan) or mouse anti-GFAP (1:1000, Bio-Rad, US). This was followed by incubation with fluorochrome-conjugated secondary antibodies against rabbit (Alexa Fluor555, Thermo Fisher, US) or mouse (Alexa Fluor488, Thermo Fisher, US) at room temperature for 1 hour. Sections were mounted with Pro-Long Gold Antifade reagent (Thermo Fisher, US) and imaged using confocal laser scanning microscope Leica TCS SP8 (Leica Microsystems, Germany).

Western blot

On POD 7 and 14, fresh C7 spinal cord segment was collected and homogenized in lysis buffer (Neuronal Protein Extraction Reagent, Thermo Fisher, US) containing protease and phosphatase inhibitor cocktail (10%, Pierce Protease and Phosphatase Inhibitor Mini Tablet, Thermo Fisher, US). Total protein concentration was determined using Pierce bicinchoninic acid (BCA) protein kit (Thermo Fisher, US). Protein was resolved in 15% sodium dodecyl sulfate (SDS) polyacrylamide-gel electrophoresis and transferred to polyvinylidene difluoride membrane (0.45 μ m) (Roche Applied Science, Germany). After blocking with 5% milk, the membranes were incubated over night with primary antibodies against Iba-1 (1:400, Wako, Japan), Stat3 (1:1,000, Cell Signalling Technology, US), phospho-Stat3 (1:1,000, Cell Signalling Technology, US) or tubulin (1:3,000, Cell Signalling Technology, US). The membranes were incubated with horseradish peroxidase conjugated secondary antibodies and visualized using the enhanced chemiluminescence detection system (Immobilon, Merck, US).

Statistics

Data were presented as mean $\pm$ standard error of the mean. For mechanical allodynia result, two-way analysis of variance (ANOVA) was performed to compare between sham and SNL or vehicle groups, and Student *t* test was used to compare between SNL or vehicle and SNL or JSKL groups at each time point. For Western blot assessment,

one-way ANOVA followed by Tukey's multiple comparison test was used. A *p* value less than .05 was considered statistically significant (two-tailed). Graph Pad Prism software was used for statistical analysis.

Result

JSKL administration attenuated SNL-induced mechanical allodynia

Rats receiving the sham operation did not present hypersensitivity after surgery compared with baseline level (Fig. 2). As compared with the sham operation group, SNL caused significant mechanical allodynia of the ipsilateral forepaw of rats of SNL or vehicle group on POD 7 and 14 ($P=.0016$ and $<.0001$ respectively, two-way ANOVA). Notably, JSKL administration significantly improved allodynia symptom on POD 7 ($P=.0210$, Student's *t* test) and 14 ($P=.0039$, Student's *t* test) compared with PBS treatment.

JSKL inhibited the activation of microglia but not astrocytes in rat spinal cord

Glia activation plays a vital role in neuropathic pain. Staining of Iba-1 (marker of microglia) and GFAP (marker of astrocytes) were applied. As expected, dramatic increase of spinal Iba-1 signal after SNL surgery was observed on POD 7 and 14 (Fig. 3). The JSKL administration significantly attenuated SNL-induced microglia activation as compared with PBS (Fig. 3A). This was further confirmed by Western blot analysis (Fig. 3B). Compared with PBS, JSKL caused a 26% ($P=.0481$) and 77% ($P=.0117$) reduction in spinal Iba-1 expression on POD 7 and 14, respectively (Fig. 3C, one-way ANOVA). Spinal astrocytes were also activated by SNL surgery as indicated by up-regulated GFAP signal

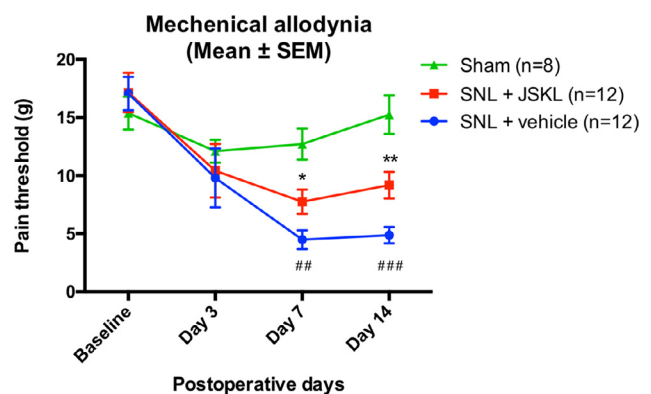


Fig. 2. Behavior testing result shown as mechanical withdrawal threshold of forepaw of rats from three groups. Mechanical allodynia induced by ligation of C7 spinal nerve was most significant on POD 7 and 14 as compared with sham group (### $p<.01$, ### $p<.001$, two-way ANOVA). Rats from SNL or JSKL group presented significantly attenuated mechanical allodynia on POD 7 and 14 as compared with rats with SNL receiving vehicle administration (* $p<.05$, ** $p<.01$, *t* test).

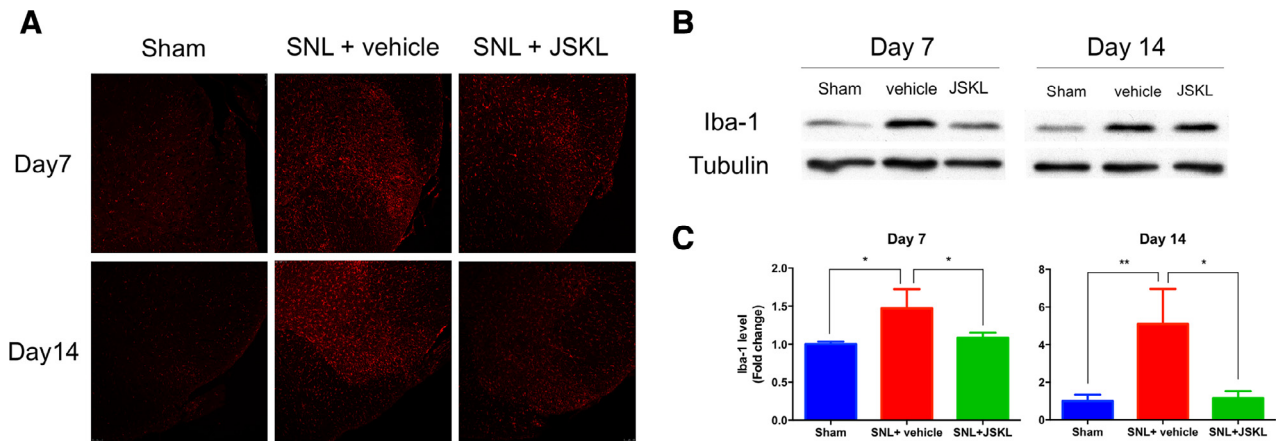


Fig. 3. Administration of JSKL decreased activation of Iba-1 positive microglia in the spinal cord. Representative immunofluorescence staining (A) showed the signal intensity of Iba-1 positive microglia in ipsilateral dorsal horn of C7 spinal cord was attenuated by JSKL administration both on POD 7 and 14 (Magnification, $\times 10$). This was further confirmed by semiquantitative analysis using Western blot (B and C), with the finding that JSKL significantly reduced SNL-induced upregulation of Iba-1 in the spinal cord on POD 7 (-39%) and 14 (-31%). (* $p < .05$, ** $p < .01$, one-way ANOVA, $n = 3$).

intensity. However, this was not reversed by JSKL administration (Fig. 4).

JSKL suppressed phosphorylation of spinal Stat-3 after nerve ligation

Previous studies showed that Stat3 contributed to microglia reactivation by regulating cell proliferation and expression of proinflammatory cytokines [19,25]. Therefore, we investigated the potential suppressive effect of JSKL on Stat3. Immunoblotting (Fig. 5) showed that the expression of spinal phosphorylated Stat3 was significantly higher in rats with SNL surgery (SNL or vehicle and SNL or JSKL) than sham surgery on POD 7 (125% higher, $P = .0005$) and 14 (236% higher, $P = .0025$, one-way ANOVA). In comparison with rats with SNL and treated with PBS, JSKL administration strongly reduced the phosphorylation of Stat3 in

the spinal cord on POD 7 (-39% , $p = .0030$) and 14 (-31% , $p = .0888$, one-way ANOVA), indicating that JSKL might inhibit spinal Stat3 activation induced by peripheral nerve injury.

Discussion

To our knowledge, our study is the first to illustrate the underlying cellular and molecular mechanism of the analgesic effect of JSKL, a traditional Chinese medicine formula, in treating cervical radicular pain. As hypothesized, we confirmed that JSKL significantly attenuated SNL-induced allodynia in a rat model of cervical radiculopathy, which was accompanied by a decreased activation of microglia in the spinal cord. More importantly, phosphorylated-Stat3 was also decreased at the same time, suggesting Stat3 might be the relevant mediator of JSKL-induced analgesic effect through downregulating microglia activation.

Radiating extremity pain is one of the most common clinical symptoms associated with cervical radiculopathy [1]. Although controlling pain with pharmaceutical regimen remains one of the major treatment strategies, definite efficacy of current pharmacological treatment is not clear [26]. This may be, at least partially, associated with the complexity of the pathophysiology of neuropathic pain [27]. Therefore, patients have been seeking alternative treatments including traditional Chinese medicine for pain control with reported positive efficacy [28–30]. In fact, a report by the World Health Organisation estimated that 80% of world's population are using herbal medicine in their daily standard health care system [31]. JSKL is one of traditional Chinese medicine formulas that has been historically prescribed in China for treatment of cervical radiculopathy, with reported efficacy in attenuating clinical cervical radicular pain [14]. However, the pharmacological mechanism of JSKL is still undefined.

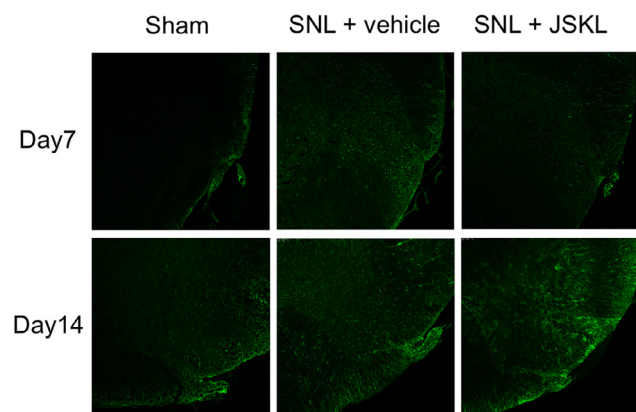


Fig. 4. GFAP positive astrocytes were not affected by JSKL administration in the spinal cord. Immunofluorescent staining showed that ligation of C7 spinal nerve resulted in hyperintensity of GFAP-positive astrocytes in the ipsilateral spinal cord both on POD 7 and 14, whereas the up-regulated signal intensity of GFAP was not affected by JSKL treatment (Magnification, $\times 10$).

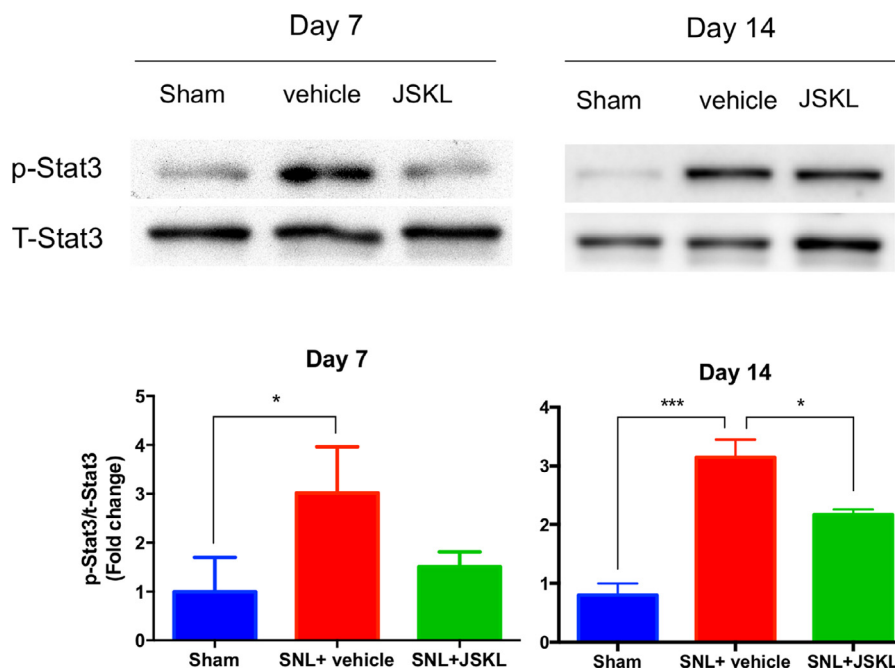


Fig. 5. Phosphorylation of Stat3 in the spinal cord after SNL was inhibited by JSKL treatment. Western blot analysis showed that phosphorylated Stat3 on tyrosine 705 (p-Stat3) was elevated in animals with SNL surgery receiving vehicle administration both on POD 7 and 14. This activation of Stat3 was significantly suppressed by JSKL administration on POD 7. (* $p < .01$, *** $p < .001$, one-way ANOVA, $n = 3$).

In our present study using a rat model of cervical SNL, daily treatment with JSKL significantly attenuated spinal nerve ligation-induced allodynia on POD 7 and 14. Notably, this analgesic effect was accompanied with a dramatically suppressed activation of Iba-1 in the spinal cord dorsal horn (Fig. 3). As the tissue-resident macrophages in central nerve system, microglia plays an important role in immune surveillance of external environment, and is extensively involved in cellular signalling transduction in response to neurological stimulus [25,32]. It was found that compression of peripheral nerve might cause microglia reactivation as indicated by greatly enhanced Iba-1 immunoreactivity (IR) in spinal cord dorsal horn [17]. Microglia reactivation led to cell proliferation and enhanced secretion of many pronociceptive mediators, such as IL-1b, brain-derived neurotrophic factor (BDNF) and cathepsin S [33–35]. These mediators in turn prompted nociceptive signal transmission of second-order neurons in spinal dorsal horn, and thus contributed to pain hypersensitivity [19,36]. On the other hand, medications blocking microglia or macrophage activation showed promising results in treating pain in both preclinical and clinic settings [17,37,38]. Our finding that JSKL administration decreased Iba-1 expression suggested microglia is one of the major targets of JSKL. This inhibition effect might be related to the active component cinnamaldehyde, which was proved to suppress inflammatory response of macrophage in vitro [39]. It could be expected that by combining cinnamaldehyde with other active components, JSKL effectively prevents microglia reactivation and thus mitigates chronic neuropathic pain. Further studies are needed to validate this

advantage of using formula over single compound in terms of pain management.

To understand the underlying mechanism of JSKL-induced microglia inhibition, we detected the phosphorylation of Stat3 protein in spinal dorsal horn. It is characterized that Stat3 is involved in microglia activation in response to external stimulus by promoting the microglia proliferation and transcription of pain modulating genes [19,25,40]. In addition, inhibition of Stat3 pathway was found sufficient to suppress cytokines release from microglia after exposure of radiofrequency fields [32]. Dominguez and coworkers also reported that phosphorylated Stat3 (p-Stat3) inhibitor was able to ameliorate neuropathic pain in a rat model of peripheral nerve injury [19]. In the current study, we found that cervical SNL-induced activation of spinal microglia was in coincidence with increased level of p-Stat3. This was in agreement with previous finding that p-Stat3 was elevated in the activated microglia in a rat model of lumbar SNL [19]. Similar to Iba-1 results, we observed that p-Stat3 expression in the spinal cord was strongly suppressed by daily JSKL administration. In fact, ferulic acid and cinnamaldehyde, two major fractions of JSKL, were reported to attenuate Stat3 activation in vitro [15,16]. Together with these evidence, we therefore postulate that JSKL may exert its analgesic affect by blocking Stat3 phosphorylation and thus preventing microglial reactivation. It will be of interests to investigate the synergic effects of several potential microglia inhibitors in future studies.

To investigate potential dose-dependent effect of JSKL on its bioactivities may provide additional information on

the mechanism of actions as well as therapeutic range for treatment of relevant patients. In our study, when JSKL was administrated at a regimen of 400 mg/day/rat (equivalent to 300 mg/kg/day in human), the SNL-induced radicular pain was significantly alleviated after 7 and 14 days of treatment. However, a lower dose (100 or 200 mg/day) failed to exert its analgesic effect noted in our pilot study, suggesting there might exist a dose-dependent effect of JSKL in treating pain via regulating Stat3 pathway. Therefore, it is important to further study the dosing effect of JSKL and its active fractions in pain management both in vivo and clinical trials with the hope of finding an optimized regimen. In addition, studies on the chemical components of JSKL may provide novel pharmacologic molecular candidates for microglia inhibition.

In addition to microglia, astrocyte is another well-accepted type of glial cell that plays an essential role in the development and maintenance of neuropathic pain [41]. As expected, SNL surgery caused great increase in GFAP-IR in spinal dorsal horn (Fig. 4). However, no difference in GFAP-IR was observed between SNL or PBS and SNL or JSKL groups. It is therefore unlikely that JSKL attenuates neuropathic pain via modulating astrocytes.

Conclusion

In summary, JSKL was found to significantly decrease mechanical allodynia in an animal model mimicking cervical radiculopathy. Oral administration of JSKL prevented the activation of pain-causing microglia, but not astrocytes in the spinal pain pathway. A coincident reduction of Stat3 phosphorylation by JSKL was also observed in spinal dorsal horn. The JSKL might interfere with Stat3-microglia pathway to attenuate neuropathic pain. Although the above findings suggest that JSKL could serve as an alternative medicine to control cervical radicular pain, further controlled clinical trials in humans is needed.

References

- [1] Carette S, Fehlings MG. Cervical radiculopathy. *N Engl J Med* 2005;353:392–9. doi: [10.1056/NEJMc043887](https://doi.org/10.1056/NEJMc043887).
- [2] Woods BI, Hilibrand AS. Cervical radiculopathy: epidemiology, etiology, diagnosis, and treatment. *J Spinal Disord Tech* 2015;28:E251–9. doi: [10.1097/BSD.0000000000000284](https://doi.org/10.1097/BSD.0000000000000284).
- [3] DeLeo JA, Hashizume H, Rutkowski MD, Weinstein JN. Cyclooxygenase-2 inhibitor SC-236 attenuates mechanical allodynia following nerve root injury in rats. *J Orthop Res* 2000;18:977–82. doi: [10.1002/jor.1100180618](https://doi.org/10.1002/jor.1100180618).
- [4] Sekiguchi M, Sekiguchi Y, Konno S, Kobayashi H, Homma Y, Kikuchi S. Comparison of neuropathic pain and neuronal apoptosis following nerve root or spinal nerve compression. *Eur Spine J* 2009;18:1978–85. doi: [10.1007/s00586-009-1064-z](https://doi.org/10.1007/s00586-009-1064-z).
- [5] Slotkin JR, Ness JK, Snyder KM, Skiles AA, Woodard EJ, O'Shea T, et al. Sustained local release of methylprednisolone from a thiol-acrylate poly(ethylene glycol) hydrogel for treating chronic compressive radicular pain. *Spine* 2015;1. doi: [10.1097/BRS.0000000000001309](https://doi.org/10.1097/BRS.0000000000001309).
- [6] Liu D-L, Wang X, Chu W-G, Lu N, Han W-J, Du Y-K, et al. Chronic cervical radiculopathic pain is associated with increased excitability and hyperpolarization-activated current (I_h) in large-diameter dorsal root ganglion neurons. *Mol Pain* 2017;13:1744806917707127. doi: [10.1177/1744806917707127](https://doi.org/10.1177/1744806917707127).
- [7] Rosenberg GJ, Yee AJM, Erwin WM. Bedside to bench and back to bedside: translational implications of targeted intervertebral disc therapeutics. *J Orthop Transl* 2017;10:18–27. doi: [10.1016/j.jot.2017.03.008](https://doi.org/10.1016/j.jot.2017.03.008).
- [8] Qin L, Yao D, Zheng L, Liu W-C, Liu Z, Lei M, et al. Phytomolecule icaritin incorporated PLGA/TCP scaffold for steroid-associated osteonecrosis: proof-of-concept for prevention of hip joint collapse in bipedal emus and mechanistic study in quadrupedal rabbits. *Biomaterials* 2015;59:125–43. doi: [10.1016/j.biomaterials.2015.04.038](https://doi.org/10.1016/j.biomaterials.2015.04.038).
- [9] Deng W, Zhang P, Huang H, Shen Y, Yang Q, Cui W, et al. Five-year follow-up study of a kidney-tonifying herbal Fufang for prevention of postmenopausal osteoporosis and fragility fractures. *J Bone Miner Metab* 2012;30:517–24. doi: [10.1007/s00774-012-0351-7](https://doi.org/10.1007/s00774-012-0351-7).
- [10] Zhu HM, Qin L, Garner P, Genant HK, Zhang G, Dai K, et al. The first multicenter and randomized clinical trial of herbal Fufang for treatment of postmenopausal osteoporosis. *Osteoporos Int* 2012;23:1317–27. doi: [10.1007/s00198-011-1577-2](https://doi.org/10.1007/s00198-011-1577-2).
- [11] Chen S, Zheng L, Xie X, Wang X, Lai Y, Chen S, et al. Comparative study of poly (lactic-co-glycolic acid)/tricalcium phosphate scaffolds incorporated or coated with osteogenic growth factors for enhancement of bone regeneration. *J Orthop Transl* 2014;2:91–104. doi: [10.1016/j.jot.2014.01.002](https://doi.org/10.1016/j.jot.2014.01.002).
- [12] Xie X-H, Wang X-L, Yang H-L, Zhao D-W, Qin L. Steroid-associated osteonecrosis: epidemiology, pathophysiology, animal model, prevention, and potential treatments (an overview). *J Orthop Transl* 2015;3:58–70. doi: [10.1016/j.jot.2014.12.002](https://doi.org/10.1016/j.jot.2014.12.002).
- [13] Li Z-R, Cheng L-M, Wang K-Z, Yang N-P, Yang S-H, He W, et al. Herbal Fufang Xian Ling Gu Bao prevents corticosteroid-induced osteonecrosis of the femoral head—a first multicentre, randomised, double-blind, placebo-controlled clinical trial. *J Orthop Transl* 2018;12:36–44. doi: [10.1016/j.jot.2017.11.001](https://doi.org/10.1016/j.jot.2017.11.001).
- [14] Lu X. The effect of Jingshukeli combined with computer inter-frequency therapy for patients with cervical spondylot-ic radiculopathy. *Chin J Pract Nerv Dis* 2015;10:35–6. doi: [10.3969/j.issn.1673-5110.2015.10.021](https://doi.org/10.3969/j.issn.1673-5110.2015.10.021).
- [15] Tsai C-M, Yen G-C, Sun F-M, Yang S-F, Weng C-J. Assessment of the anti-invasion potential and mechanism of select cinnamic acid derivatives on human lung adenocarcinoma cells. *Mol Pharm* 2013;10:1890–900. doi: [10.1021/mp3006648](https://doi.org/10.1021/mp3006648).
- [16] Huang J-S, Lee Y-H, Chuang L-Y, Guh J-Y, Hwang J-Y. Cinnamaldehyde and nitric oxide attenuate advanced glycation end products-induced the JAK/STAT signaling in human renal tubular cells. *J Cell Biochem* 2015;116:1028–38. doi: [10.1002/jcb.25058](https://doi.org/10.1002/jcb.25058).
- [17] Yao Y, Tan Y-H, Light AR, Mao J, Yu ACH, Fu K-Y. Alendronate attenuates spinal microglial activation and neuropathic pain. *J Pain* 2016;17:889–903. doi: [10.1016/j.jpain.2016.03.008](https://doi.org/10.1016/j.jpain.2016.03.008).
- [18] Hubbard RD, Chen Z, Winkelstein BA. Transient cervical nerve root compression modulates pain: load thresholds for allodynia and sustained changes in spinal neuropeptide expression. *J Biomech* 2008;41:677–85. doi: [10.1016/j.jbiomech.2007.09.026](https://doi.org/10.1016/j.jbiomech.2007.09.026).
- [19] Dominguez E, Rivat C, Pommier B, Mauborgne A, Pohl M. JAK/STAT3 pathway is activated in spinal cord microglia after peripheral nerve injury and contributes to neuropathic pain development in rat. *J Neurochem* 2008;107:50–60. doi: [10.1111/j.1471-4159.2008.05566.x](https://doi.org/10.1111/j.1471-4159.2008.05566.x).
- [20] Nair AB, Jacob S. A simple practice guide for dose conversion between animals and human. *J Basic Clin Pharm* 2016;7:27–31. doi: [10.4103/0976-0105.177703](https://doi.org/10.4103/0976-0105.177703).
- [21] Wu J, Mo J, Guo Y, Liang H. Observation of clinical curative effect of Jingshu granule in treatment of nerve root cervical spondylosis of Qi stagnation and blood stasis. *China Med Pharm* 2017;7:41–4.
- [22] Jihan H, Xiaozuo H, Zhiyang C, Qingshan Z, Ruiyuan S. Dose conversion among different animals and healthy volunteers in pharmacological study. *Chin J Clin Pharmacol Ther* 2004;9:1069–72.

- [23] Liu X, Tian Y, Lu N, Gin T, Cheng CHK, Chan MTV. Stat3 inhibition attenuates mechanical allodynia through transcriptional regulation of chemokine expression in spinal astrocytes. *PLoS One* 2013;8: e75804. doi: [10.1371/journal.pone.0075804](https://doi.org/10.1371/journal.pone.0075804).
- [24] Dixon WJ. The up-and-down method for small samples. *J Am Stat Assoc* 1965;60:967–78. doi: [10.1080/01621459.1965.10480843](https://doi.org/10.1080/01621459.1965.10480843).
- [25] Przanowski P, Dabrowski M, Ellert-Miklaszewska A, Kloss M, Mieczkowski J, Kaza B, et al. The signal transducers Stat1 and Stat3 and their novel target Jmjd3 drive the expression of inflammatory genes in microglia. *J Mol Med* 2014;92:239–54. doi: [10.1007/s00109-013-1090-5](https://doi.org/10.1007/s00109-013-1090-5).
- [26] Bono CM, Ghiselli G, Gilbert TJ, Kreiner DS, Reitman C, Summers JT, et al. An evidence-based clinical guideline for the diagnosis and treatment of cervical radiculopathy from degenerative disorders. *Spine J* 2011;11:64–72. doi: [10.1016/j.spinee.2010.10.023](https://doi.org/10.1016/j.spinee.2010.10.023).
- [27] Zhang Y, Wang C, Wang L, Parks GS, Zhang X, Guo Z, et al. A novel analgesic isolated from a traditional chinese medicine. *Curr Biol* 2014;24:117–23. doi: [10.1016/j.cub.2013.11.039](https://doi.org/10.1016/j.cub.2013.11.039).
- [28] Ingram SL. Pain: novel analgesics from traditional chinese medicines. *Curr Biol* 2014;24:R114–6. doi: [10.1016/j.cub.2013.12.030](https://doi.org/10.1016/j.cub.2013.12.030).
- [29] Lee K-H. Research and future trends in the pharmaceutical development of medicinal herbs from Chinese medicine. *Public Health Nutr* 2000;3:515–22. doi: [10.1017/S1368980000000604](https://doi.org/10.1017/S1368980000000604).
- [30] Zhu L, Gao J, Yu J, Feng M, Li J, Wang S, et al. Jingtong granule: a chinese patent medicine for cervical radiculopathy. *Evid-Based Complement Altern Med* 2015;2015:e158453 2015. doi: [10.1155/2015/158453](https://doi.org/10.1155/2015/158453). 2015.
- [31] Chan K. Some aspects of toxic contaminants in herbal medicines. *Chemosphere* 2003;52:1361–71. doi: [10.1016/S0045-6535\(03\)00471-5](https://doi.org/10.1016/S0045-6535(03)00471-5).
- [32] Lu Y, He M, Zhang Y, Xu S, Zhang L, He Y, et al. Differential pro-inflammatory responses of astrocytes and microglia involve STAT3 activation in response to 1800 MHz radiofrequency fields. *PLoS One* 2014;9:e108318. doi: [10.1371/journal.pone.0108318](https://doi.org/10.1371/journal.pone.0108318).
- [33] Coull JAM, Beggs S, Boudreau D, Boivin D, Tsuda M, Inoue K, et al. BDNF from microglia causes the shift in neuronal anion gradient underlying neuropathic pain. *Nature* 2005;438:1017–21. doi: [10.1038/nature04223](https://doi.org/10.1038/nature04223).
- [34] Clark AK, Yip PK, Grist J, Gentry C, Staniland AA, Marchand F, et al. Inhibition of spinal microglial cathepsin S for the reversal of neuropathic pain. *Proc Natl Acad Sci* 2007;104:10655–60. doi: [10.1073/pnas.0610811104](https://doi.org/10.1073/pnas.0610811104).
- [35] Hanisch U-K. Microglia as a source and target of cytokines. *Glia* 2002;40:140–55. doi: [10.1002/glia.10161](https://doi.org/10.1002/glia.10161).
- [36] Liu T, van Rooijen N, Tracey DJ. Depletion of macrophages reduces axonal degeneration and hyperalgesia following nerve injury. *Pain* 2000;86:25–32. doi: [10.1016/S0304-3959\(99\)00306-1](https://doi.org/10.1016/S0304-3959(99)00306-1).
- [37] Ellis A, Wieseler J, Favret J, Johnson KW, Rice KC, Maier SF, et al. Systemic administration of propentofylline, ibudilast, and (+)-naltrexone each reverses mechanical allodynia in a novel rat model of central neuropathic pain. *J Pain Off J Am Pain Soc* 2014;15:407–21. doi: [10.1016/j.jpain.2013.12.007](https://doi.org/10.1016/j.jpain.2013.12.007).
- [38] Sevcik MA, Luger NM, Mach DB, Sabino MAC, Peters CM, Ghilardi JR, et al. Bone cancer pain: the effects of the bisphosphonate alendronate on pain, skeletal remodeling, tumor growth and tumor necrosis. *Pain* 2004;111:169–80. doi: [10.1016/j.pain.2004.06.015](https://doi.org/10.1016/j.pain.2004.06.015).
- [39] Chao LK, Hua K-F, Hsu H-Y, Cheng S-S, Lin I-F, Chen C-J, et al. Cinnamaldehyde inhibits pro-inflammatory cytokines secretion from monocytes/macrophages through suppression of intracellular signaling. *Food Chem Toxicol* 2008;46:220–31. doi: [10.1016/j.fct.2007.07.016](https://doi.org/10.1016/j.fct.2007.07.016).
- [40] Huang C, Ma R, Sun S, Wei G, Fang Y, Liu R, et al. JAK2-STAT3 signaling pathway mediates thrombin-induced proinflammatory actions of microglia in vitro. *J Neuroimmunol* 2008;204:118–25. doi: [10.1016/j.jneuroim.2008.07.004](https://doi.org/10.1016/j.jneuroim.2008.07.004).
- [41] Tanga FY, Raghavendra V, DeLeo JA. Quantitative real-time RT-PCR assessment of spinal microglial and astrocytic activation markers in a rat model of neuropathic pain. *Neurochem Int* 2004;45:397–407. doi: [10.1016/j.neuint.2003.06.002](https://doi.org/10.1016/j.neuint.2003.06.002).